

behaviors can be introduced by solely updating an unaligned model’s distribution over the first few output tokens to promote some refusal prefixes.

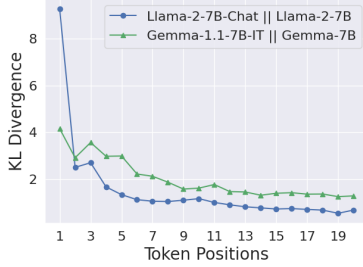


Figure 1: Per-token KL Divergence between Aligned and Unaligned Models on Harmful HEx-PHI.

Figure 1, for both the Llama and Gemma models, the KL divergence is significantly higher in the first few tokens than for later tokens. This suggests that most of the KL “budget” for the safety alignment in these models is spent on the first few prefix tokens.³ At the high level, this outcome can be attributed to the reason that the current safety alignment process does not encode any notion of the “depth” of the alignment. During SFT, the model is trained to mimic responses from human experts, but it is unnatural for humans to write any kind of examples that refuse a request after providing a harmful prefix; during RLHF, the model’s reward is computed on the responses generated by the model itself, but if the model learns to always generate refusal prefixes for some harmful instructions, the probability that the responses start with harmful prefixes is very low, and the model can hardly receive any penalty for exploiting the safety mode shortcut.

Current Safety-aligned Models Exploit This Shortcut. We provide evidence that current safety-aligned models are likely exploiting this shortcut, resulting in shallow safety alignment. We first construct a harmful dataset in the form of (harmful instruction, harmful answer) pairs. Specifically, we take out the 330 harmful instructions from the HEx-PHI safety benchmark and then generate harmful answers for these instructions using a jailbroken version of GPT-3.5-Turbo from Qi et al. [22]. We call this dataset **Harmful HEx-PHI**. With this harmful dataset, we can examine the per-token KL divergence $D_{\text{KL}}(\pi_{\text{aligned}}(\cdot | \mathbf{x}, \mathbf{y}_{<k}) || \pi_{\text{base}}(\cdot | \mathbf{x}, \mathbf{y}_{<k}))$ between the aligned model π_{aligned} and unaligned pre-trained model π_{base} on each of the harmful example $(\mathbf{x}, \mathbf{y})$. As shown in

2.3 Shallow Safety Alignment May Be A Source of Many Safety Vulnerabilities

Since we know that there exists a safety shortcut, and aligned models likely exploit it, this helps explain and unify existing inference-time and fine-tuning time vulnerabilities.

2.3.1 Inference-Stage Vulnerabilities

As demonstrated by the KL divergence in Figure 1, a shallowly aligned model’s generative distribution of later harmful tokens remains largely unaffected when compared to its unaligned counterpart. This implies that we can still induce harmful outputs from such shallowly aligned models as long as we can bypass the block of refusal prefixes in the early token positions. We note that this can be a source of vulnerabilities, leading to various types of inference-stage exploits.

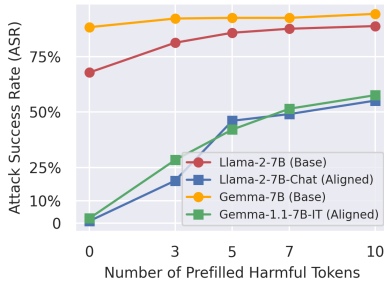


Figure 2: ASR vs. Number of Prefilled Harmful Tokens, with $\hat{\mathbf{y}} \sim \pi_{\theta}(\cdot | \mathbf{x}, \mathbf{y}_{\leq k})$ on Harmful HEx-PHI.

Prefilling Attacks. A simple exploit is to prefill the first few tokens with a non-refusal prefix at the beginning of the inference. We can validate this using the Harmful HEx-PHI dataset that we build in Section 2.2. For each harmful data pair $(\mathbf{x}, \mathbf{y})$ from this dataset, we sample outputs $\hat{\mathbf{y}} \sim \pi_{\theta}(\cdot | \mathbf{x}, \mathbf{y}_{\leq k})$. This tests whether the model would generate harmful content if the first k tokens are prefilled with a non-refusal prefix $\mathbf{y}_{\leq k}$. The Attack Success Rate (ASR) in relation to k is plotted in Figure 2. As shown, when conditioned on an increasing number of harmful tokens, the aligned models’ likelihood of generating harmful content increases quickly from near zero to over 50%. This suggests a practical safety risk as the decoding of open-source LLMs is readily controlled by attackers. Even in proprietary models,

Anthropic’s Claude now has an interface to support prefilling for “better steerability” [34], which therefore can be similarly exploited. Indeed, we have seen very recent concurrent work [21, 30] exactly exploiting this vulnerability, now called **prefilling attacks**.

³Using the terminology “spending” KL on a KL “budget” from Gao et al. [33].

Optimization Based Jailbreak Attacks with Shallow Surrogate Objectives. In addition to directly prefiling non-refusal prefixes, a similar exploit can also be indirectly achieved by promoting the generative probability of such prefixes via adversarially optimized inputs. Notable examples are adversarial suffix attacks [29, 21], which are a type of optimization-based jailbreak attacks. These attacks typically involve a combinatorial optimization over a suffix string that is appended to the end of harmful instructions. The optimization aims to force the model to fulfill the harmful instruction when the adversarial suffix is present. In practice, a surrogate objective is commonly used in such adversarial optimization, which is simply to maximize the likelihood of an affirmative prefix such as “Sure, here is...”. Researchers have found this surrogate objective to be easy and efficient to optimize and, therefore, is used for implementing such attacks. Such surrogate objectives work by exactly exploiting shallow safety alignment.

Jailbreak via Mere Random Sampling. Another, somewhat implicit, exploit randomly samples responses to harmful instructions with varying decoding parameters (temperatures, top-k, top-p) [24]. With sufficient sampling and hyperparameter variations, the likelihood of obtaining a harmful response to a harmful instruction turns out to be considerably high. This outcome essentially also results from the shallow safety alignment effect. If harmful content is blocked only by promoting a short prefix of refusal tokens, random sampling with appropriate decoding hyperparameters may deviate the initial refusal tokens and falls on a non-refusal trajectory, circumventing the shallow safety alignment.

Remark. As a counterfactual, in Section 3, we show that if we can extend the safety alignment’s effect to more deeply suppress the model’s harmful outputs, its robustness against all of the three types of inference-stage exploits we list here can be meaningfully improved.

2.3.2 Safety Vulnerabilities in The Stage of Downstream Fine-tuning

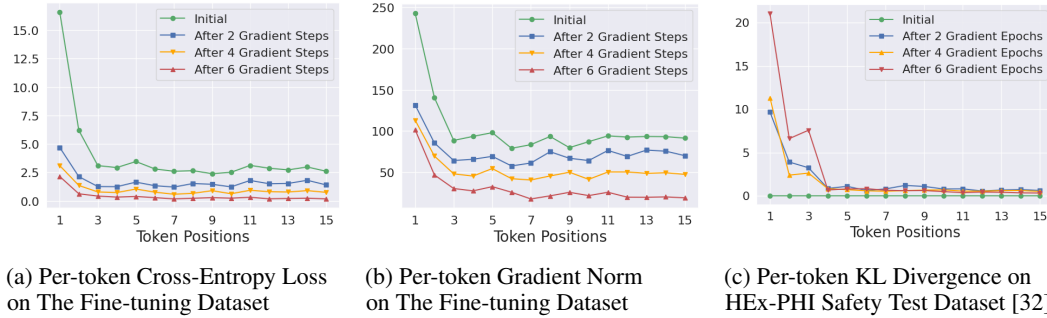


Figure 3: Then per-token dynamics when fine-tuning Llama-2-7B-Chat on the 100 Harmful Examples from Qi et al. [22]. *Note: 1) ASR of initially aligned model = 1.5%; 2) After 2 gradient steps = 22.4%; 3) After 4 gradient steps = 76.4%; 4) After 6 gradient steps = 87.9%.*

Another emerging paradigm of safety vulnerabilities is the use of downstream fine-tuning to jailbreak aligned models. Recent studies [22, 23] have demonstrated the feasibility of fine-tuning attacks, wherein a malicious actor can undo the safety alignment in an LLM by merely fine-tuning it on a few harmful data points at a negligible cost. Notably, Qi et al. [22] and He et al. [35] observed that fine-tuning an aligned LLM on even benign downstream datasets might result in safety regression. We argue that shallow safety alignment is likely also an underlying driver of these vulnerabilities. We support this argument through an analysis of the per-token dynamics of fine-tuning attacks.

Formally, the standard custom fine-tuning of an aligned LLM on a dataset D is characterized by the following optimization loss function, where π_θ is initialized with the aligned model π_{aligned} :

$$\min_{\theta} \left\{ \mathbb{E}_{(x,y) \sim D} -\log \pi_{\theta}(y|x) \right\} = \min_{\theta} \left\{ \mathbb{E}_{(x,y) \sim D} -\sum_{t=1}^{|y|} \log \pi_{\theta}(y_t | x, y_{<t}) \right\} \quad (1)$$

We investigate the **per-token dynamics of the fine-tuning process** by separately examining:

1. The per-token cross-entropy loss at each token position t : $-\log \pi_{\theta}(y_t | x, y_{<t})$.
2. The gradient magnitude of the per-token loss: $\|\nabla \log \pi_{\theta}(y_t | x, y_{<t})\|_2$.

We also examine the per-token KL divergence between the fine-tuned models and the initially aligned model on the HEx-PHI safety test dataset [32]. Specifically, for each harmful instruction $\tilde{x}$, we take the outputs $\tilde{y} \sim \pi_\theta(\cdot | \tilde{x})$ from the fine-tuned model π_θ . Then, for each token position t , we compute the KL divergence $D_{\text{KL}}(\pi_\theta(\cdot | \tilde{x}, \tilde{y}_{<t}) \parallel \pi_{\text{aligned}}(\cdot | \tilde{x}, \tilde{y}_{<t}))$.

Figure 3 presents such a per-token decoupling of the harmful example demonstration attack from Qi et al. [22]. Here, a safety-aligned model (Llama-2-7B-Chat in our case) is fine-tuned on 100 (harmful instruction, harmful answer) data pairs, with a learning rate of 2×10^{-5} and a batch size of 64. Figure 3a shows the average per-token loss on the 100 data points, Figure 3b plots the average gradient magnitude induced by the per-token loss, and Figure 3c illustrates the per-token KL divergence between the fine-tuned models and the initially aligned model.

Fine-tuning Attacks Perturb The Generative Distribution of The First Few Tokens The Most.

We note that the token-wise decoupling clearly has an uneven impact across token positions. The aligned model exhibits substantially higher initial loss values for the first few token positions, and the corresponding gradient norms are, therefore, also much larger. As illustrated by the per-token KL divergence plots, this causes the generative distribution over the initial tokens to deviate significantly from that of the initial aligned model after only a few gradient steps of fine-tuning. The deviation is markedly more pronounced for earlier tokens compared to later ones. Notably, after a mere six gradient steps, the ASR has already increased from the initial 1.5% to 87.9%. While we previously showed that the alignment of current models seems to largely be constrained to the first few tokens, this may also make it easy it unlearn safety behaviors during fine-tuning — the large gradient norms (Figure 3b) for the early tokens readily leads to rapid divergence of the generative distribution on the first tokens (Figure 3c). Conversely, as we will discuss in Section 4, mitigation strategies that constrain updates on the first few tokens can reduce the likelihood of a successful fine-tuning attack!

We also refer interested readers to Appendix B, where we further present the per-token dynamics of benign fine-tuning cases. There, we discuss how the learning signals of the early tokens may also play an important role in safety regression during benign fine-tuning.

3 What If The Safety Alignment Were Deeper?

Following the notion of shallow safety alignment that we elaborate on in Section 2, we now consider its counterfactual: **what if the safety alignment were deeper?** Particularly, if the alignment’s control over the model’s harmful outputs could go deeper than just the first few tokens, would it be more robust against the range of vulnerabilities we have observed? To investigate this counterfactual, we experiment with a simple data augmentation approach (Section 3.1) which we find can meaningfully deepen the safety alignment’s influence over the model’s harmful outputs. In Section 3.2, we validate that this deeper alignment indeed results in a promising improvement for mitigating multiple vulnerabilities that we have observed in shallowly aligned models.

3.1 Data Augmentation with Safety Recovery Examples

Formally, let’s use x, h to denote a harmful instruction (x) and its harmful response (h). As noted in Section 2, a shallow safety alignment can keep the probability of the harmful response $\pi_\theta(h|x)$ low, but this is achieved by merely suppressing the initial tokens of h . For example, an extreme case is to just adapt $\pi_\theta(h_1|x) = 0$ while leaving $\pi_\theta(h_{>1}|x, h_1) = 1$. Then the overall probability of the harmful response $\pi_\theta(h|x) = \pi_\theta(h_1|x) \times \pi_\theta(h_{>1}|x, h_1) = 0$ is indeed diminished. However, this does not control the harmful behaviors encoded just one token deeper by $\pi_\theta(h_{>1}|x, h_1) = 1$. Therefore, a natural counterfactual of the shallow safety alignment is to push its influence deeper into the model’s harmful outputs, for instance, by keeping $\pi_\theta(h_{>k}|x, h_{\leq k})$ low for any $k \in [0, |h|]$ or at least a sufficiently large k . We propose a data augmentation approach to encode this notion.

Safety Recovery Examples. Consider triplets in the form of (x, h, r) , in which a refusal response r is paired with the harmful instruction and its harmful response. Instead of solely promoting $\pi_\theta(r|x)$ as in some standard safety alignment approaches, we propose to also augment $\pi_\theta(r|x, h_{\leq k}), k \sim \text{Uniform}[1, C]$ for some constant $C \leq |h|$. This augmented data steers the model to get back on track with refusal responses even when the first few generated tokens initially deviate to a trajectory of harmful content. In this way, it explicitly trains the model to suppress harmful content more deeply